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A Scale-Coupled Method for Simulation of the Formation and Evolution of Aeolian Dune Field

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Abstract

In this paper, a scale-coupled method for the formation and evolution of aeolian sand dune fields is proposed. It successfully reproduces various dune patterns and the whole development process of a dune field of several square kilometers during several decades. Based on our simulations, we investigate the influence of sand supply, wind velocity, and wind direction to the dune patterns and dune speed in the dune field in quantity. Compared with the existing dune field model, we solve the correspondence problem in length-scale and time-scale between the simulated and the actual formation and evolution of a dune field. Our results on the variations of the dune speed with the dune height and the relation between the thickness of sand supply and the average dune height both agree with the results measured in field in quality and in quantity. Besides, we present the influence of wind velocity and sand diameter to the maximum dune height in a dune field and replay the 'solitary wave'-like behavior of two barchan dunes in a dune field. These results are not only beneficial to understand the dynamical characteristic of a dune field more comprehensively and more clearly, but also meaningful for further prediction of the migration trend of deserts under global climate change and the investigation of the aeolian geomorphology on Mars.

Keywords: Scale-coupled method; Aeolian sand dune field; 'Solitary wave'-like behavior; Aeolian Geomorphology

1 Introduction

Being cognizant of intense global climate change and vagarious wind-formed landforms on Mars, more and more scientists pay their attention to the study of sand dunes, especially to the formation and evolution of dune field [1-3]. Besides the long-term field observations [4-5] and experimental researches [6-7], some numerical calculation methods based on the continuum theory, and discrete dynamical methods used for computer simulations were proposed to reveal the characteristic of an aeolian sand dune field. In the former models, the incoming wind flow and the wind-blown sand flow were generally treated as a steady incompressible boundary-layer flow and a fluid-like granular flow respectively. Through solving the corresponding differential equations [8], sand flux at every position of the initial configuration was calculated to determine the rate of the change of bed elevation [9], and then the time evolution of the bed surface was given by the mass conversation. It's worth noting that in this type of models, an initial bed configuration which was similar to the pattern of a sand dune, such as a cone, was assigned as the initial condition. Their resulted pattern was close to the actual topography, such as barchan dunes, whose height depended on the height of the initial configuration [10]. In the latter's models, the initial bed surface was discretized

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into many cells with identical cross-section areas but different heights, such as 'lattice columns' whose heights are equal to the local thickness of sand supply [11] or 'sand slabs' whose heights are identically taken as 1/3m [12]. The transportation of these discrete cells were artificially regulated, for example, to move along the wind direction with a distance determined by the relative height of two neighboring 'lattice columns' [11], or to move 5 meters directly [12], and the deposition rule was associated with the height of 'lattice columns' [11] or the deposition probability given by Bagnold [12]. For a randomly given 3-D initial configuration, the formation and evolution of dune field can be replayed through repeatedly operating these rules by computer.

The superiority of the numerical calculation methods based on the continuum theory is that it has a sound understanding of the physical mechanisms behind dune formation and migration which therefore can realize the corresponding relationship in length-scale and time-scale between the simulated results and the actual situations, the simulated results are strictly derived from an accurate calculation of flow field and the scale of initial configuration. However, for a dune field where a large number of dunes with different sizes and shapes coexist, an accurate calculation of the flow field becomes a difficult task, which makes this method hard to be applied in the study of the dynamical characteristics of the whole dune field. Although the existing discrete dynamical methods can replay the formation and evolution of the whole dune field, but the operation rules in these computer simulations are generally arbitrarily assigned, which are deficient in reflecting the actual wind-blown sand movement process and the influence of some physical quantities, such as wind velocity and sand flux. However, these processes and quantities play a key role in the formation and evolution of a dune field. Therefore, their simulation results seem similar to the actual dune field in pattern, but poorly agree with the formation and evolution of actual dune field in length-scale and time-scale. In fact, the fundamental task in simulating the formation and evolution of dune field is to solve the trans-scale problem, that is, the transition from the small scale physical processes (such as the transport of sand particles) to the large scale physical processes (such as the formation and evolution of dune fields). Obviously, such trans-scale transition is not a simple linear superposition. In other words, it is impossible to characterize the formation and evolution of a dune field by modeling the process of every sand particle, which brings an infinite number of degrees of freedom into the modeling. Apart from capability limitation of present computers, the characteristics and physical mechanism of the formation and evolution of dune field are inherently different from those for sand movement and sand ripples. However, the dynamical behaviors of large-scale dune fields are definitely related with sand motions. Therefore, ignorance or complete parameterization of sand motions would possibly cause mis-reflecting the real formation and evolution physical process of dune field.

In this paper, we incorporate the advantages of the numerical calculation methods and the discrete dynamical methods introduced above, and propose a scale-coupled method to simulate the formation and evolution of a dune field. In our model, a conceptual material element– 'sand slab' is employed as a 'transitional element'. It bridges the saltation movements of sand particles which is micrometers in length-scale and seconds in time-scale, and the formation and evolution of a dune field which is kilometers in length-scale and hundred years in time-scale. Incorporating the influence of local topography on the intensity of wind field and sand flux, erosion and deposition, this model successfully solves the correspondence problem between the simulated and the actual formation and evolution of dune field in length-scale and time-scale, and the trans-scale problem from microscopic sand movement to macroscopic formation and evolution of dune field. The simulated results agree well with the observational results in quality and in quantity. It reveals the effects of friction wind velocity and sand diameter on dune fields. Besides, it replays a 'solitary wave'-like behavior occurring in a dune field.

2 Simulation method

The basic idea of the scale-coupled method for the simulation of sand dune field is to mesh the sand bed into rectangular sand slabs, and define both the microscopic and macroscopic transportation properties and erosion-deposition behavior of 'sand slabs'. On the microscopic level, the thicknesses of 'sand slabs' are firstly determined by analyzing the physical process in wind-blown sand flow, then the transportation length of 'sand slabs' are determined based on the mechanism of entrainment, rebound and deposition of sand particles with the aid of the probability and statistics theory. On the macroscopic level, the positions of eroded 'sand slabs' as well as the deposition of 'sand slabs' are determined by analyzing the deformation of sand bed. The major procedure of the scale-coupled method of sand dune field is described as following.

Step 1: The discretization of sand bed. Different from previous models that initial sand heaps or an initial morphology of sand bed were needed, in our model, for a given thickness of sand supply H_s, sand diameter D_s and simulation area, a flat sand bed is divided into many rectangular units with length a and width b, respectively, called 'sand slabs'. The positions of 'sand slabs' can be represented by the subscript i and j. For example, for a simulation area of 3000m×3000 m and $a = b = 1$m, we get $i = 1, 2\dots, 3000$ and $j = 1, 2\dots 3000$.

Step 2: Determine the real friction wind velocity $u_{*ij}^{(n)}$ in the nth day according to the shape of sand bed. In previous dune field models, the influence of wind velocity on the eroded sand flux and the transportation length was neglected. In order to remedy this deficiency, it is necessary to calculate the variations of the friction wind velocity in the whole dune field. Generally speaking, at the windward slope of a sand dune, the acceleration increment of the downstream friction wind velocity is linear to the height of sand bed $h_{ij}^{(n-1)}$ [13]. Inspired by the experimental results on wind acceleration given by Lancaster [14], we can determine the friction wind velocity at any position in any time interval $\Delta T = T/N$ as follow:

$$u_{*ij}^{(n)} = u_{*-\infty} + \Delta u_{*ij}^{(n)} \tag{1}$$

where $\mathrm{D}u_{*ij}^{(n)}$ can be expressed as a function of the slope angle $q_{ij}^{(n-1)}$ and surface elevation $h_{ij}^{(n-1)}$:

$$\Delta u_{*ij}^{(n)} = \frac{h_{ij}^{(n)}}{h_{ijtop}^{(n-1)}}[u_{*1top}^{(n-1)} - u_{*-\infty} - \left(u_{*1top}^{(n-1)} - u_{*2top}^{(n-1)}\right) \frac{\tan\alpha_1^{(n-1)} - \tan\varphi_{ij}^{(n-1)}}{\tan\alpha_1^{(n-1)} - \tan\alpha_2^{(n-1)}}] \quad n=2,3,\dots \tag{2}$$

in which:

$$\varphi_{ij}^{(n-1)} = \alpha_1^{(n-1)} + \ln\left[\frac{h_{ijtop}^{(n-1)} - h_{1top}^{(n-1)}}{\left(h_{2top}^{(n-1)} - h_{1top}^{(n-1)}\right)e^{\alpha_1^{(n-1)} - \alpha_2^{(n-1)}}} + \frac{h_{2top}^{(n-1)} - h_{ijtop}^{(n-1)}}{h_{2top}^{(n-1)} - h_{1top}^{(n-1)}}\right] \tag{3}$$

is related to the elevations $h_{1top}^{(n-1)}$, $h_{2top}^{(n-1)}$ and the windward slope angles $\theta_1^{(n-1)}$, $\theta_2^{(n-1)}$ of any two dunes on the (n−1)th day. Initially, that is, when $n=1$, the sand bed is flat, $\Delta u_{*ij}^{(1)} = 0$ and $u_{*top}^{(1)} = u_{-\infty} + 0.01 h_{ijtop}^{(1)}$. In addition, in the protection area (which is an area ranging from the top of sand dune to the recovery point of wind. Its length is about four times of the dune height and the angle is normally 15 degree), the friction wind velocity is set as zero.

Step 3: Determine the position of the eroded 'sand slab' and the sand transportation rate $q_{sij}^{(n)}$ at this position. In previous dune field models, the determination of eroded position didn't involve the influence of friction wind velocity. For example, in Werner's model [12], eroded positions were selected randomly from the sand bed; while in Momiji's model [11], all positions in the sand bed were eroded in each time step. In our model, the positions of eroded 'sand slabs' are determined by the local friction wind velocity, that is, by calculating the friction wind velocity $u_{*ij}^{(n)}$ at any position, we can determine the eroded positions according to whether the friction wind velocity $u_{*ij}^{(n)}$ at this position exceeds the threshold value u_{*t}. After that, considering the variations of the configuration of sand bed, we need to identify the sand transportation rate $q_{sij}^{(n)}$ at each eroded position. Based on the continuum saltation model[9], in which the saltation layer of wind-blown sand flow was treated as fluid-like granular, we can derive the sand flux $q_{sij}^{(n)}$ and steady state velocity $u_{sij}^{(n)}$ of wind-blown sand flow at different positions through solving the mass and momentum conservation equation. The resulted expressions can be written as:

$$q_{sij}^{(n)} = 2\gamma u_{sij}^{(n)}[(u_{*ij}^{(n)})^2 - u_{*t}^2]/g \tag{4}$$

And

$$u_{sij}^{(n)} = [2u_{*ij}^{(n)}\sqrt{z_1/z_m + (1 - z_1/z_m)u_{*t}^2/(u_{*ij}^{(n)})^2} + u_{*t}\ln(z_1/z_m) - \kappa]/\kappa - \sqrt{\rho_{sand}Dg/3C_d\gamma} \tag{5}$$

where $\kappa = 0.4$ is the Von Karman constant; $g = 9.81\text{m}\cdot\text{s}^{-2}$ is the gravity acceleration; $\rho_{sand} = 2650\text{kg}\cdot\text{m}^{-3}$ is the density of sand; $z_m = 0.04$ m is the thickness of saltation layer; $z_1 = 0.005$ m is the reference height, and $\gamma = 0.4$ is the model parameter; C_d is the drag coefficient.

Step 4: Determine the thicknesses $\eta_{ij}^{(n)}$ of every eroded 'sand slabs'. In previous dune field models, the eroded sand flux at every position was treated as a constant or a function of the gradient of bed elevation. While in our model, the influences of friction wind velocity and sand diameter are incorporated into the determination of eroded sand flux at every position based on the saltation transport process in the wind-blown sand flow. For a given time step $\Delta T = 3600 \times 24$s, that is one day, after obtaining the sand flux eroded during ΔT, that is, $Q_{sij}^{(n)} = q_{sij}^{(n)}\Delta T b(1 - \Delta s)$, we can deduce the average thickness of the eroded 'sand slabs': $\eta_{ij}^{(n)} = Q_{sij}^{(n)}/(\sigma\rho_s S_{ij})$, in which σ is the porosity of sand bed and usually is supposed to be 0.62; Δs is the coverage coefficient which is defined as the ratio between the length of the sand area covered by the deposited sands after one-time saltating and the average saltation length $L_{sij}^{(n)} = u_{sij}^{(n)}T_{sij}^{(n)}$, that is:

$$\Delta s \approx q_{sij}^{(n)}T_{sij}^{(n)}(\beta + \alpha)/(\beta\rho_{sand}DL_{sij}^{(n)}) \tag{6}$$

where α and β are the deposition and erosion rate of sand particles; $T_{sij}^{(n)} = 1.7u_{*ij}^{(n)}/g$ is the average saltation time [15]; $n_t = |\Delta T / T_{sij}^{(n)}|$ is mean saltation times of sands within ΔT; $|\cdot|$ is the integral function.

Step 5: Randomly select one of the eroded 'sand slabs', and calculate its transportation length $l_{ij}^{(n)}$. Similarly, the transportation length of the eroded 'sand slab' in previous dune filed model was also treated as a constant or a function of the gradient of bed elevation. While in our model, it depends on the local friction wind velocity and sand diameter. Firstly, we assume the mass of the 'sand slab' in site (i, j), $M_{ij}^{(n)} = a \times b \times B_{ij}^{(n)} \times \rho_s$, is eroded for n_t times during ΔT, and the time required for each erosion is $T_{sij}^{(n)}$; sand mass eroded each time is $M_{ij}^{(n)}/n_t$; Considering the momentum conservation of wind-blown sand flow at steady state, and the sum of the average momentum of the sand deposited into the area with unit width and $n_t L_{sij}^{(n)}$ in length, namely, $\sum_{n'=1}^{n} m_{ij}^{(n')} v_{ij}^{(n')}$, is supposed to be equal to the momentum of the 'sand slab', $M_{ij}^{(n)} l_{ij}^{(n)} / \Delta T$, therefore, we can get the average transportation length $l_{ij}^{(n)}$ of the sand slab S_{ij} by

$$l_{ij}^{(n)} = \sum_{n'=1}^{n_t} m_{ij}^{(n')} \cdot v_{ij}^{(n')} \frac{\Delta T}{M_{ij}^{(n)}} = \sum_{n'=1}^{n_t} m_{ij}^{(n')} \cdot n' \cdot \frac{L_{sij}^{(n)}}{M_{ij}^{(n)}} \tag{7}$$

where $m_{ij}^{(n')}$ represents the whole sand mass deposited at location $(i + n'L_{sij}^{(n)}, j)$ which can be expressed as:

$$m_{ij}^{(n')} = \sum_{j'=1}^{n_t+1-n'} \Delta M_{j'}\left(i + n'L_{sij}^{(n)}, j\right) \tag{8}$$

where $\Delta M_{j'}(i+n'L_{sij}^{(n)},j)$ represents the sand mass that leaves from (i, j) and deposits at $(i+n'L_{sij}^{(n)},j)$ at the j'th time. It can be deduced as follows: First, the sand mass that deposits at $(i+L_{sij}^{(n)},j)$ can be expressed as

$$\Delta M_{j'}\left(i+L_{sij}^{(n)},j\right)=\begin{cases}\dfrac{M_{ij}^{(n)}}{n_t}\alpha(1-\beta)^{n_t-(j'+1)} & j'=1,\ldots,n_t-1\\ \Delta M_{n_t}(i+L_{sij}^{(n)},j)=\dfrac{M_{ij}^{(n)}}{n_t} & j'=n_t\end{cases} \quad (9)$$

Since the sand mass that leaves from $(i+L_{sij}^{(n)},j)$ and deposits at $(i+2L_{sij}^{(n)},j)$ can be determined from the sand mass that leaves from (i, j) and deposits at $(i+L_{sij}^{(n)},j)$ from the $(j'+1)$th time to the n_tth time, we get:

$$\begin{aligned}\Delta M_{j'}\left(i+2L_{sij}^{(n)},j\right)=&(1-\alpha)\Delta M_{j'+1}\left(i+L_{sij}^{(n)},j\right)+\\&\sum_{k=2}^{n_t-j'}\left[\alpha(1-\beta)^{k-2}-\alpha(1-\beta)^{k-1}\right]\\&\cdot\Delta M_{j'+k}\left(i+L_{sij}^{(n)},j\right)\quad j'=1,\ldots,n_t-1\end{aligned} \quad (10)$$

Accordingly, the sand mass that leaves from $(i+L_{sij}^{(n)},j)$ and deposits at $(i+n'L_{sij}^{(n)},j)$ can be determined from the sand mass that leaves from (i, j) and deposits at $(i+(n'-1)L_{sij}^{(n)},j)$ from the $(j'+1)$th time to the (n_t+2-n')th time:

$$\begin{aligned}\Delta M_{j'}\left(i+n'L_{sij}^{(n)},j\right)=&(1-\alpha)\Delta M_{j'+1}\left[i+(n'-1)L_{sij}^{(n)},j\right]+\\&\sum_{k=2}^{n_t+2-n'-j'}\left[\alpha(1-\beta)^{k-2}-\alpha(1-\beta)^{k-1}\right]\\&\cdot\Delta M_{j'+k}\left[i+(n'-1)L_{sij}^{(n)},j\right]\quad j'=1,\ldots,n_t+1-n'\end{aligned} \quad (11)$$

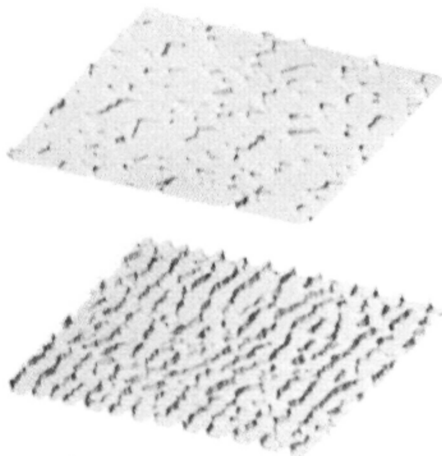
It is worth noting that, in natural wind-blown sand flow, the property of sand bed is heterogeneous which results in the inconsistency of the deposition probability α in the whole field. For simplicity, we randomly let some positions are of deposition probability α' according to a certain proportion, and the transportation lengths of the corresponding 'sand slabs' in these positions are identically set as $l_{ij}^{(n)}/2$. Besides, if a 'sand slab' is transported into the protection area, we let it deposit directly.

Step 6: Modification of the configuration of sand bed. Due to the collapse behavior of sand, we need to set the maximum gradient of adjacent positions to be same as the angle of repose Θ (generally taken as 30°), after the eroded 'sand slab' is transported and deposited. That means, if the gradient of sand bed is larger than Θ then the 'sand slab' at the higher position moves to the adjacent lower position. The thickness of the 'sand slab' moved is taken as $(\Delta h-\sqrt{S_{ij}}\tan\Theta)/2$, in which Δh is the height difference of two adjacent positions.

After that, another eroded 'sand slab' within time ΔT is randomly selected to repeat step 5 and 6 until all of the eroded 'sand slabs' in this ΔT are treated. Then go to the treatments for the next ΔT, that is: select two different dunes in the whole field obtained in the nth ΔT, determine $u_{*1top}^{(n)}$ and $u_{*2top}^{(n)}$ according to the heights of the two dunes and their windward slope angles, then go to step 2. The whole simulation ends until the simulation time satisfies the required time T, for example, n = 100 year.

3 Simulation results and discussion

With the method introduced above and the self-compiling programs, we conducted quantitative simulations on a dune field under different conditions of thickness of sand supply H_s, sand diameter D_s and incoming wind velocity $u_{*-\infty}$. In which, the simulation area and the size of each 'sand slab' are taken as 3000×3000 m^2 and a=b=1m, respectively. Fig.1 presents the configurations of the whole dune fields in the 15th year, for $u_{*-\infty}$= 0.5m·s^{-1} and D_s= 0.3mm. The thicknesses of sand supply are taken as H_s= 1m (Fig.1a) and H_s=4m (Fig.1b), respectively. It can be found that, for H_s= 1m, the dune field is dominated by barchan dunes, while for H_s= 4m, the patterns of the sand dunes in the dune field are mainly transverse dunes. These results are consistent with the field observational results [16].



(a)

(b)

Fig.1. Configurations of the whole dune field in the 15th year, for $u_{*-\infty}$= 0.5m·s^{-1} and D_s= 0.3mm. The simulation area is 3000×3000 m^2, and the thicknesses of sand supply are taken as (a) H_s=1m and (b) H_s=4m respectively.

Due to the fact that the existing field observations and theoretical models haven't provided the influence of sand diameter and strength of wind field to the dune height in a dune field, Fig.2 illustrates the variations of the maximum dune height h_{max} of sand dunes versus the sand diameter D_s (Fig.2a) and the incoming friction wind velocity $u_{*-\infty}$ (Fig.2b), the thicknesses of sand supply are taken as 1m and 4m, respectively. From Fig.2, we can find that for the same conditions of wind velocity, wind direction and thickness of sand supply, the increase of sand diameter leads to the decrease of the maximum dune height; while for the same conditions of thickness of sand supply and sand diameter, the maximum dune height increases with the friction wind velocity increasing. In addition, with time passing, the increase rate of the maximum dune height becomes smaller, but the change tendency of the maximum dune height with wind velocity and sand diameter doesn't vary. The change law of the average dune height h_{ave} in a dune field with the thickness of sand supply is qualitatively and quantitatively consistent with field observation results, as shown in Fig.3. For example, when the thickness of sand supply is 2m, the predicted average dune height is 8.35m which agrees with the observational result, 8m given by ref. [17] (for $u_{*-\infty}$= 0.5m·s^{-1} and D_s= 0.3mm).

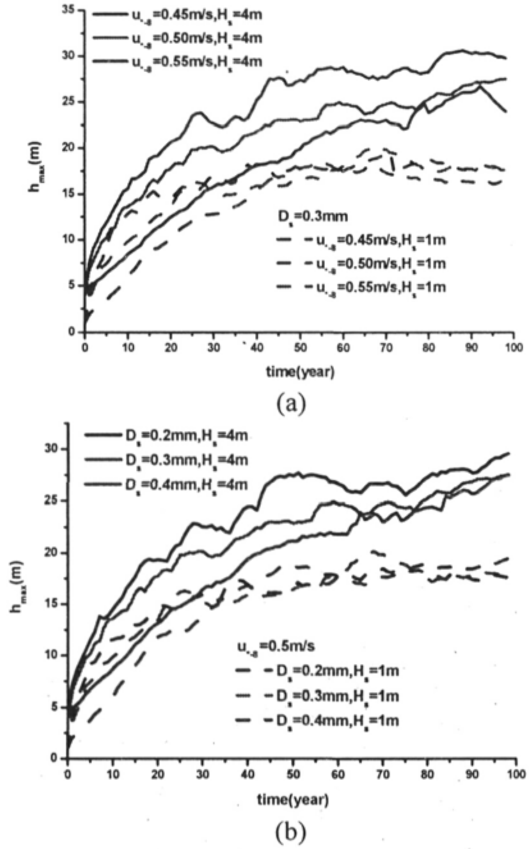


Fig.2. Variations of the maximum dune height (h_{max}) in the dune field for (a) different friction wind velocities and (b) sand diameters

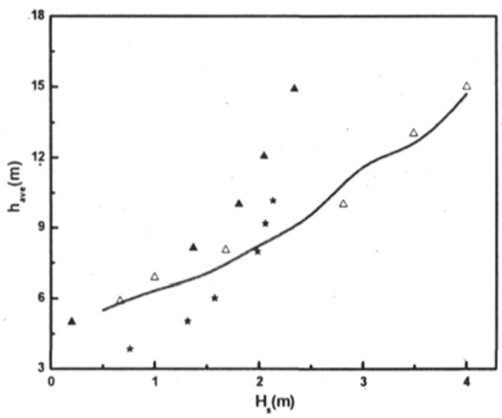


Fig.3. Comparisons between the results of the field observation and simulation results on the average dune height (h_{ave}) versus the thickness of sand supply, ▲, △ and ⋆ are the measured results in the Great sandy Desert, in the Simpson-Strzelecki Desert, and in the Great Voctoria, respectively [17]; —— the simulation results for $u_{*-\infty}$=0.55 m·s^{-1}, D_s=0.3 mm; the simulation area is 3000m×3000 m.

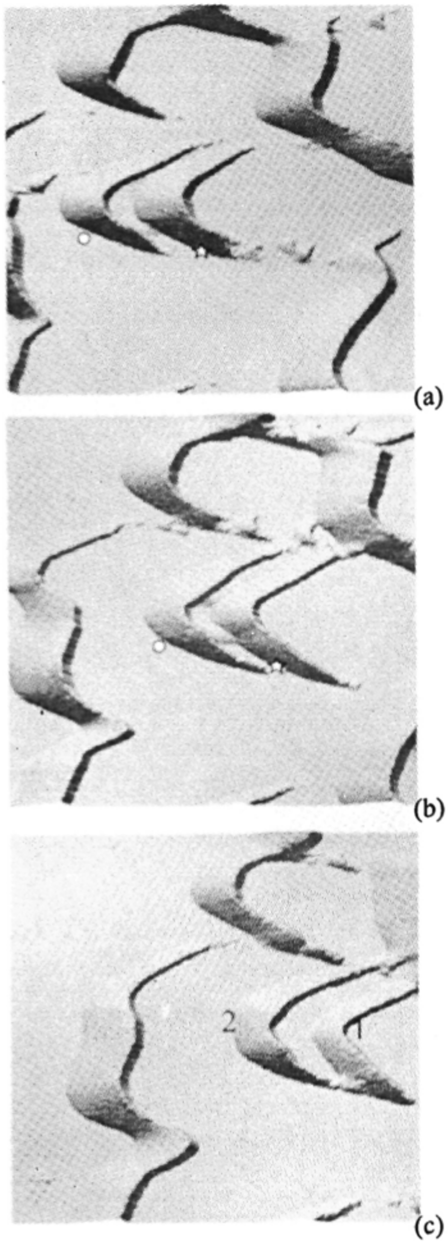


Fig.4. The 'solitary wave'-like behavior observed in the simulation with H_s=1m, D_s=0.3mm and $u_{*-\infty}$= 0.5m·s^{-1}; (a) the smaller dune (marked by '○', 8.6m in height) bumps into the larger one (marked by '□', 10.4m in height) in the 32nd year; (b) the smaller dune becomes bigger and therefore slower, while the larger one loses its mass in the 34.5th year and (c) the two dunes depart from the collision state in the 37th year, the previously smaller dune (denoted by '2') now becomes 8.9m in height, while the previously bigger one (denoted by '1') becomes 7.6m in height.

With the numerical calculation method introduced above, Schwammle and Herrmann [10] replayed the 'solitary wave' behavior of two colliding dunes successfully, that is, two loose-grained dunes are able to pass through one another while still preserving their shape. However, field observations haven't provided any evidences for this phenomenon till now [18]. Yet, there aren't any simulation results based on the Cellular Automata (CA) model and the Coupled Mapping Lattice (CML) model can support the existence of the 'solitary wave' behavior in the dune field. In our simulated dune field for H_s= 1m, D_s= 0.3mm and $u_{*-\infty}$= 0.5m·s^{-1}, we find that (see in Fig.4), two similar-sized dunes approach each other during their migration along the wind direction (Fig.4a); the smaller dune (○) in front gradually becomes bigger and therefore slower, while the larger one (□) behind gradually loses its mass (Fig.4b); Finally the dune that was previously bigger now becomes the smaller one, and its velocity becomes sufficiently large to leave the collision state (Fig.4c). After collision, the height of the previously bigger dune (denoted by '1' in Fig.4c) is close to that of the previously smaller one (denoted by '○' in Fig.4a), while the height of the previously smaller one (denoted by '2' in Fig.4c) gets close to that of the previously smaller one (denoted by '□' in Fig.4a), which seems that the smaller dune in front (denoted by '○' in Fig.4a) crosses the bigger one (denoted by '□' in Fig.4a), whereas in reality the two dunes never merge, owing to mass exchange. In the view of this paper, this phenomenon of dune interactions observed in our simulation can be deemed as some kind of 'solitary wave'-like behavior. The reason for this phenomenon is that, the smaller dune in front moves with a relative large velocity which catches up with the larger dune behind, during their interaction, the slip face of the dune in front plays a shadow effect to the dune behind which results in a part of the sands in the windward slope of the dune behind not being eroded and then gained by the dune in front as two dunes moving. Therefore, the previously bigger dune becomes the smaller one, and the previously smaller one becomes larger. Through our simulations for different friction wind velocities, sand diameters and thicknesses of sand supply, we find that centered collisions between two dunes are rare to be observed, let alone the centered collisions between two similar-sized dunes. That is why on large-scale dune fields this phenomenon has not been observed till now. On the other hand, since the previous discrete dynamical methods, no matter the CA model [12] or the CML model [11], didn't consider the actual wind-blown sand transportation and deposition process, it is difficult for them to predict this phenomenon. Of course, the 'solitary wave'-like behaviors are expected to be further discussed.

4. Conclusion

In this paper, a scale-coupled method is proposed to simulate the formation and evolution of a whole dune field. Compared with the previous numerical calculation methods and discrete dynamical methods, it doesn't need an initial heap for a sand dune or an initial morphology for a dune field. In addition, this model incorporates the variations of wind field and sand transport intensity with the topography of a dune field into determining the positions and thickness of eroded 'sand slabs', and establishes a relationship between the average saltation length and the transportation length of 'sand slab'. The simulated results successfully present the rules and characteristics of dune fields which agree well with the observational results. At the same time, it reveals the change law of the dune height in a dune field with sand diameter and thickness of sand supply which can't be predicted by the previous models. The method proposed in this paper solves not only the correspondence problem between the simulated and the actual formation and evolution of dune field in length-scale and time-scale, but also the trans-scale problem from microscopic sand movement to macroscopic formation and evolution of a dune field. Simulation results show that the thickness of sand supply has a significant influence on patterns of sand dunes in a dune field. For an insufficient sand supply, the dune field is dominated by barchan dunes; while for a sufficient sand supply, the dune field is dominated by transverse dunes. The results on the variations of dune height with thickness of sand supply agree with the observational results in quality and in quantity. For the same conditions of wind velocity, wind direction and thickness of sand supply, the increase of sand diameter leads to the decrease of the maximum dune height; while for the same conditions of thickness of sand supply and sand diameter, the maximum dune height increases with the friction wind velocity increasing. With time passing, the increase rate of the maximum dune height becomes smaller. Besides, through our simulations, we find that with the dune field developing, it is possible for the 'solitary wave'-like behavior occurring between two similar-sized dunes in a barchan dune field.

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